Bayesian Analysis on Kozu Obsidian Synthetic Data

June 23, 2026

1 Overview

In this PDF, I highlight the basic idea of Bayesian analysis based on the online lecture: Statistical Rethinking (`https://github.com/rmcelreath/stat_rethinking_2026`) by Richard McElreath. The typical Bayesian analysis follows the core workflow shown in Fig. 1. Since this pioneer project does not have its own data, the analysis was made by the synthetic data initialized by my personal assumptions.

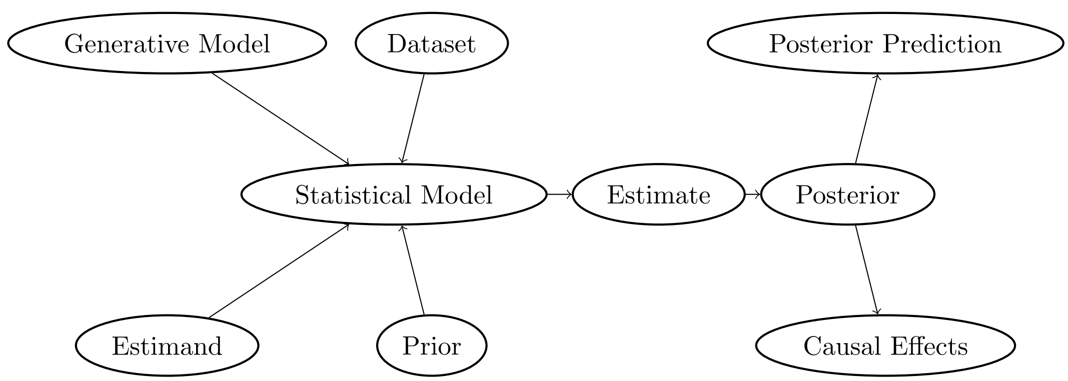

Figure 1: Core Bayesian Workflow, proposed in 305

The parameters in the dataset forms causal relationship in Fig. 2, based on my assumption. Publication and publication year can bias the proportion of Kozu obsidians. Although S, O and K can be related each other, I included single causality to be simple.

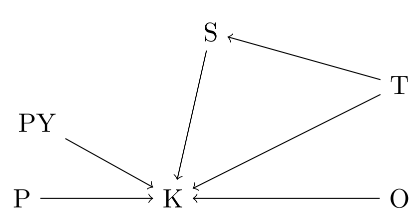

Figure 2: DAG of causalities among attributes of dataset - K: a ratio of Kouzu obsidians, S: a ratio of Shinshu obsidians, O: other sites, T: typology of artifacts or associated artifacts in the same layer, P: publication, PY: publication year.

Now, hypothesis of the analysis is sea level and proportion of Shinshu obsidians affect the Kozu obsidians to check the degree of influence. The parameters are T, K, S, dK, and dS. Y, SL, and U are hidden and do not exist in the data. U is the standard deviation of the model.

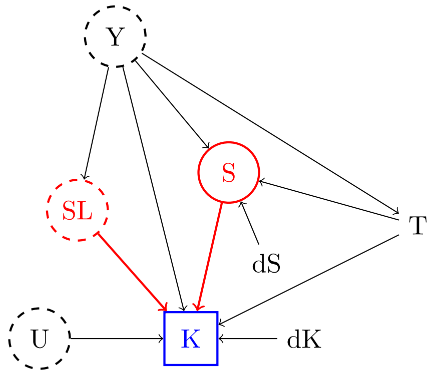

Figure 3: DAG of causalities among parameters - K: a ratio of Kouzu obsidians (blue solid square), S: a ratio of Shinshu obsidians (red dashed circle), SL: sea level difference based on current level (m; red dashed circle), T: typology of artifacts or associated artifacts in the same layer, dK: distance from Kozusima (km), dS: distance from the centre of Shinshu production sites (km), Y: true date (dashed circle as it is hidden), U: other hidden parameters influencing K (dashed). Colored arrows are from predictors to estimate.

2 Three Models Proposed here

As a background knowledge, we know the association of SL and Y based on previous research (). Therefore, multi-level modeling is used. The basic idea is that as T is observed parameter of Y, Y can be the True value of T, in other words. Then, I can make multilevel models, including Y predicts T to simulate the true Y values. As Y is estimated in ulam model of rethinking package, we can use it to estimate K, S, and SL. The relationship between Y and T can be based on the Bayesian Analysis of Typological dates by Kanezaki, Omori, and Kobayashi in 2025 (`https://www.cambridge.org/core/journals/radiocarbon/article/optimal-bayesian-chronological-modeling-for-high-resolution-chronology-for-the-middle-jomon-of 4B1717D6D82FFA1CF8A63E6B1EBCCC8D`). However, in this case, T is modeled simple poisson distribution with 25 phases.

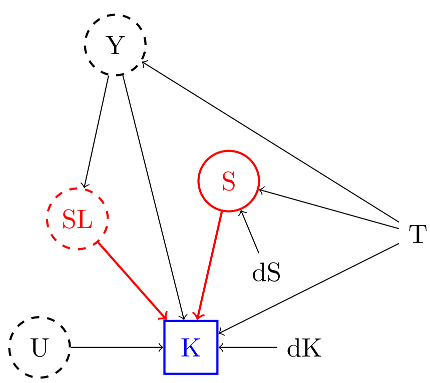

Figure 4: DAG of causalities among parameters - K: a ratio of Kouzu obsidians (blue solid square), S: a ratio of Shinshu obsidians (red dashed circle), SL: sea level difference based on current level (m; red dashed circle), T: typology of artifacts or associated artifacts in the same layer, dK: distance from Kozusima (km), dS: distance from the centre of Shinshu production sites (km), Y: true date (dashed circle as it is hidden), U: other hidden parameters influencing K (dashed). Colored arrows are from predictors to estimate.

- Model 1: stratifies data by T, and includes all the causation in Fig. 3: Model T by Y, Model SL by Y, Model S by Y and dS, Model K by S, SL, Y, dK, and U.
- Model 2: stratifies data by T, and includes all the causation in Fig. 4: Model T by Y, Model SL by Y, Model S by dS, Model K by S, SL, Y, dK, and U with correlation between S and SL.
  - Although Model 2 does not rely on Y to predict S, the causation is represented as correlation between S and SL.
- Model 3: includes all the causation in Fig. 3 without any stratification: Model T by Y, Model SL by Y, Model S by Y and dS, Model K by S, SL, Y, dK, and U.
  - See the m.Rmd and m.html

Models from next page shows the full model including all parameters; however, the analysis removes several parameters to ensure the reliability of sampling.

In this analysis, I will decide what is the best model for the synthetic data. However, it might not be the case for actual dataset.

3 The priors and statistical model for Model 1 (m1)

- This model stratifies S, SL, and Y and includes hierarchical models of S and SL by Y

- Model K

$$K \sim Normal(\mu_K, U)$$
$$logit(\mu_K) = \alpha_K[T] + \beta_{K_S}[T] \cdot S + \beta_{K_{SL}}[T] \cdot SL + \beta_{K_Y}[T] \cdot Y + \beta_{K_{dK}} * dK$$

  - We do not know anything about this but think similar for all group.
  - So use partial pooling to regularize the estimate.
  - mean should start at 0 and has small variation.
  - In this case (m1), we do not include correlation.
  - This is because multilevel models of S and SL includes the information of Y
  - But we have to regularize.

$$\alpha_K = \overline{\alpha_K} + \sigma_{\alpha_K} * z_{\alpha_K}[T]$$
$$\beta_{K_S} = \overline{\beta_{K_S}} + \sigma_{\beta_{K_S}} * z_{\beta_{K_S}}[T]$$
$$\beta_{K_S L} = \overline{\beta_{K_{SL}}} + \sigma_{\beta_{K_{SL}}} * z_{\beta_{K_{SL}}}[T]$$
$$\beta_{K_Y} = \overline{\beta_{K_Y}} + \sigma_{\beta_{K_Y}} * z_{\beta_{K_Y}}[T]$$

$$\overline{\alpha_K}, \overline{\beta_{K_S}}, \overline{\beta_{K_{SL}}}, \overline{\beta_{K_Y}} \sim Normal(0, 0.3)$$
$$\sigma_{\alpha_K}, \sigma_{\beta_{K_S}}, \sigma_{\beta_{K_{Sl}}}, \sigma_{\beta_{K_Y}} \sim Exponential(6)$$
$$z_{\alpha_K}[T] \sim Normal(0, 0.1)$$
$$z_{\beta_{K_S}}[T] \sim Normal(0, 0.1)$$
$$z_{\beta_{K_{SL}}}[T] \sim Normal(0, 0.1)$$
$$z_{\beta_{K_Y}}[T] \sim Normal(0, 0.1)$$

  - Followings don't need regularization
  - We are hoping/believing b_{dK} is negative, but this is not prior.

$$\beta_{K_{dK}} \sim Normal(0, 0.3)$$
$$U \sim Exponential(1)$$

- Model S

$$S \sim Normal(\mu_S, \sigma_S)$$
$$logit(\mu_S) = \alpha_S[T] + \beta_{S_Y} \cdot Y + \beta_{S_{dS}} * dS$$

  - I am hoping/believing β_{dS} is negative and β_{Y_S} is positive
  - partial pooling for the same reason for α_S and β_{Y_S}

$$\alpha_S = \overline{\alpha_S} + \sigma_{\alpha_S} * z_{\alpha_S}[T]$$
$$\beta_{S_Y} = \overline{\beta_{S_Y}} + \sigma_{\beta_{S_Y}} * z_{\beta_{S_Y}}[T]$$

$$\overline{\alpha_S}, \overline{\beta_{S_Y}} \sim Normal(0, 0.1)$$
$$\sigma_{\alpha_S}, \sigma_{\beta_{S_Y}} \sim Exponential(6)$$
$$z_{\alpha_S}[T] \sim Normal(0, 0.1)$$
$$z_{\beta_{S_Y}}[T]\ normal(0, 0.1)$$

  - don't need regularization

$$\beta S_{dS} \sim Normal(0, 0.3)$$

$$\sigma_S \sim Exponential(1)$$

- Model SL by Y
  - The log function and prior was refered to the previous research of Sea Level Change over time.
  - This function is abstruct so I will have to actually stan code to provide accurate estimates
  - ulam code in r does not allow us to hartd code the conditional function.
  - Also, SL has to be between 0 and 1.

$$SL \sim Normal(inv_logit(6 * Y - 2) - 0.04, 0.05)$$

- Model T by Y

$$T \sim Poisson(Y * 25)$$

  - This should be hard coded based on the previous research (Kanezaki, et al., 2025)
- Model Y randomly

$$Y \sim Normal(0.5, 0.2)$$

  - With a constraint that Y between 0 and 1.

4 The priors and statistical model for Model 2 (m2)

- This model stratifies S, SL, and Y and includes correlation between S and SL.

- Main model K

$$K \sim Normal(\mu_K, U)$$
$$logit(\mu_K) = \alpha_K[T] + \beta_{K_S}[T] \cdot S + \beta_{K_{SL}}[T] \cdot SL + \beta_{K_Y}[T] \cdot Y + \beta_{K_{dK}} * dK$$

  - We do not know anything about this but think similar for all group.
  - So use partial pooling to regularize the estimate.
  - mean should start at 0 and has small variation.
  - In this case (m2), we do include correlation between S and SL due to Y.

$$\alpha_K = \overline{\alpha_K} + \sigma_{\alpha_K} * z_{\alpha_K}[T]$$
$$\beta_{K_Y} = \overline{\beta_{K_Y}} + \sigma_{\beta_{K_Y}} * z_{\beta_{K_Y}}[T]$$
$$\overline{\alpha_K}, \overline{\beta_{K_Y}} \sim Normal(0, 0.3)$$
$$\sigma_{\alpha_K}, \sigma_{\beta_{K_Y}} \sim Exponential(1)$$
$$z_{\alpha_K}[T] \sim Normal(0, 0.1)$$
$$z_{\beta_{K_Y}}[T] \sim Normal(0, 0.1)$$

  - Correlaton between β_{K_S} and β_{K_SL}
  - This $compose_noncentered()$ is in rethinking package.
  - This is not too strong but good correlation

$$\beta_{K_S}, \beta_{K_SL} \sim NonCenteredMultiNormal(\sigma_{S,SL}, \rho_{S,SL}, z_{S,SL})$$
$$\sigma_{S,SL} \sim exponential(2)$$
$$\rho_{S,SL} \sim lkj_corr_cholesky(1)$$
$$z_{S,SL} \sim Normal(0, 1)$$

  - Followings don't need regularization
  - We are hoping/believing b_{dK} is negative, but this is not prior.

$$\beta_{K_{dK}} \sim Normal(0, 0.3)$$
$$U \sim Exponential(1)$$

- Model S
  - Belows are the same coding as M1

$$S \sim Normal(\mu_S, \sigma_S)$$
$$logit(\mu_S) = \alpha_S[T] + \beta_{S_{dS}} * dS$$
$$\alpha_S = \overline{\alpha_S} + \sigma_{\alpha_S} * z_{\alpha_S}[T]$$
$$\beta_{S_Y} = \overline{\beta_{S_Y}} + \sigma_{\beta_{S_Y}} * z_{\beta_{S_Y}}[T]$$
$$\overline{\alpha_S}, \overline{\beta_{S_Y}} \sim Normal(0, 0.1)$$
$$\sigma_{\alpha_S}, \sigma_{\beta_{S_Y}} \sim Exponential(6)$$
$$z_{\alpha_S}[T] \sim Normal(0, 0.05)$$
$$z_{\beta_{S_Y}}[T]\ normal(0, 0.05)$$
$$\beta S_{dS} \sim Normal(0, 0.1)$$
$$\sigma_S \sim Exponential(1)$$

- Model SL by Y

$$SL \sim Normal(inv_logit(6 * Y - 2) - 0.04, 0.05)$$

- Model T by Y

$$T \sim Poisson(Y * 25)$$

- Model Y randomly

$$Y \sim Normal(0.5, 0.2)$$

5 The priors and statistical model for Model 3 (m3)

- This model does not stratify any parameters, and includes T parameter.

- Model K

$$K \sim Normal(\mu_K, U)$$

$$logit(\mu_K) = \alpha_K + \beta_{K_S} \cdot S + \beta_{K_{SL}} \cdot SL + \beta_{K_Y} \cdot Y + \beta_{K_{dK}} * dK + \beta_{K_T}[T]$$

  – Basically same as m1 but not regularization (no pooling).

  – β_{K_T} is regularized to maintain the efficiency.

$$\alpha_K, \beta_{K_S}, \beta_{K_{SL}}, \beta_{K_Y}, \beta_{K_{dK}} \sim Normal(0, 0.1)$$

  – This normal distribution has to have the sigma of 0.1. Otherwise, parameter T does not move as expected.

  – Priors below also has to be strange like this.

$$\beta_{K_T} = \overline{\beta_{K_T}} + \sigma_{\beta_{K_T}} * z_{\beta_{K_T}}[T]$$

$$\overline{\beta K_T} \sim Normal(0, 0.1)$$

$$\sigma_{\beta_{K_T}} \sim Exponential(1)$$

$$z_{\beta_{K_T}}[T] \sim Normal(0, 0.1)$$

$$U \sim Exponential(1)$$

- Model S

  – Belows are similar to M1, though β_{S_T} are added.

$$S \sim Normal(\mu_S, \sigma_S)$$

$$logit(\mu_S) = \alpha_S + \beta_{S_{dS}} \cdot dS + \beta_{S_Y} \cdot Y + \beta_{S_T}[T]$$

$$\alpha_S, \beta_{S_Y}, \beta S_{dS} \sim Normal(0, 0.1)$$

$$\beta_{S_T} = \overline{\beta_{S_T}} + \sigma_{\beta_{S_T}} * z_{\beta_{S_T}}[T]$$

$$\overline{\beta S_T} \sim Normal(0, 0.1)$$

$$\sigma_{\beta_{S_T}} \sim Exponential(1)$$

$$z_{\beta_{S_T}}[T] \sim Normal(0, 0.1)$$

$$\sigma_S \sim Exponential(1)$$

- Model SL by Y

$$SL \sim Normal(inv_logit(6 * Y - 2) - 0.04, 0.05)$$

- Model T by Y

$$T \sim Poisson(Y * 25)$$

- Model Y randomly

$$Y \sim Normal(0.5, 0.2)$$